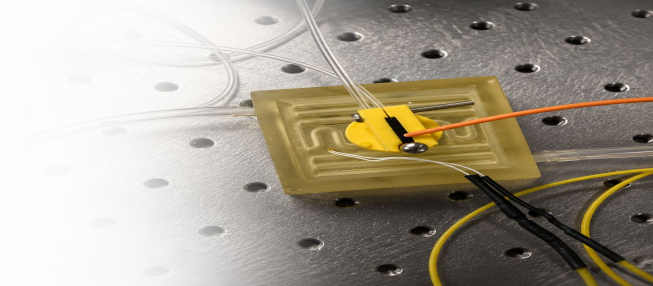


MRI-Compatible Incubation Device for Brain Organoid Imaging

[Project Source](#)

Designed and validated a temperature-controlled incubation system that preserves organoid viability during extended MRI imaging experiments



THE CHALLENGE

High-field MRI provides powerful insights into brain organoid function but current systems cannot maintain organoid viability during long imaging sessions due to lack of temperature control and nutrient delivery

OUR SOLUTION



TEMPERATURE CONTROL

Maintains ~ 37°C with water bath system



MEDIA PERFUSION

Continuous flow for nutrient delivery and waste removal



OPTICAL FIBER PORT

Integrated fiber clamp for optogenetic stimulation

KEY RESULTS



37.3 ± 0.5°C

Stable temperature maintained inside the MR tube



Continuous Perfusion

~400 μ L volume maintained with peristaltic pump flow



MRI-Compatible Design

Constructed with non-magnetic materials and optimized for MRI environment



Sustained Metabolic Activity

Metabolic assay shows organoids remain viable after 5 hours of incubation



Preserved Neuronal Activity

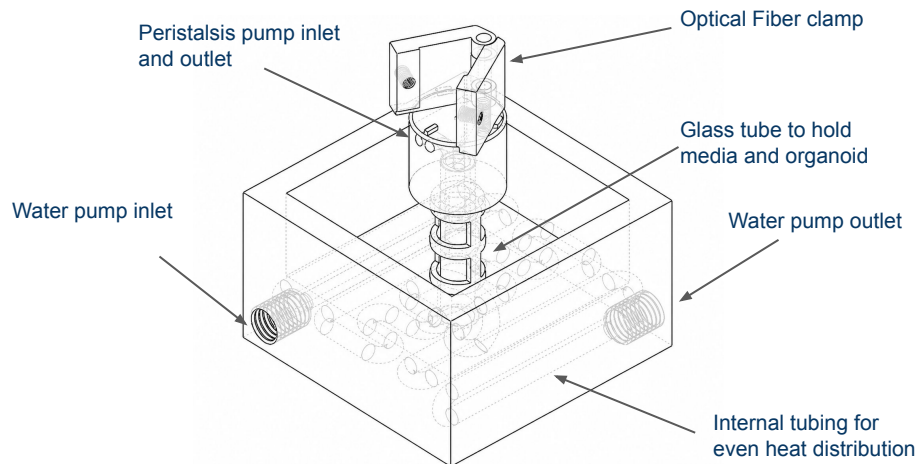
Calcium imaging confirms functional neural signaling post-incubation



Optogenetic compatibility

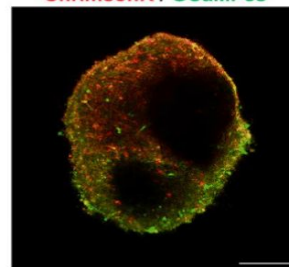
Optical fiber integrones enables precise stimulation during imaging

CAD MODEL



CALCIUM IMAGING

ChrimsonR / GCaMP8s



- CrimsonR $\Rightarrow$ Red shifted opsin
- GCaMP8s $\Rightarrow$ Calcium Indicator

Brain organoid expressing ChrimsonR and GCaMP8s. Optogenetic stimulation elicited calcium responses, indicating maintained neuronal function following incubation within the MRI-compatible chamber.